SCOPEbnb

Remote Imaging Guide

Your first night on the RedCat 91 rig at Starfront Observatories, Rockwood, Texas.

William Optics RedCat 91
448 mm f/4.9

ZWO AM5N · ASI2600MC Pro
EFW 5 x 2 in

Field of view 3.00° x 2.01°
1.73 arcsec/pixel

Software: N.I.N.A. 3.2
ASICap · RustDesk

Welcome. This guide takes you from your first remote login to a full night of data. Read it once before your session. Everything in it refers to N.I.N.A., the software already running on the observatory PC, and to ASICap, which drives the pier camera.

The rig is preconfigured. You do not need to build a sequence from scratch, calibrate anything, or connect equipment by hand. Your job for the night is: load the ready made sequence, pick a target, frame it, write it into the sequence, and press start.

> **The whole night in five lines**
>
> 1. Connect with RustDesk.
> 2. Sequencer: load **BASE_1target_L-EXTREME** or **BASE_1target_NO-FILTER**. Always first.
> 3. Sky Atlas: find a target that stays above 30 degrees for 4 hours or more.
> 4. Framing: compose and rotate, then **Update Existing Target in Sequencer**.
> 5. Check the filter, change **nothing else**, press start and go to bed.
>
> Choose the template for tonight's target: **L-EXTREME** for emission nebulae; **NO-FILTER** for broadband targets (see Section 7). Both templates are tested. Update the target's name, RA, Dec and rotation; leave timings, autofocus, meridian flip, calibration frames and shutdown unchanged.

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1. What you are controlling

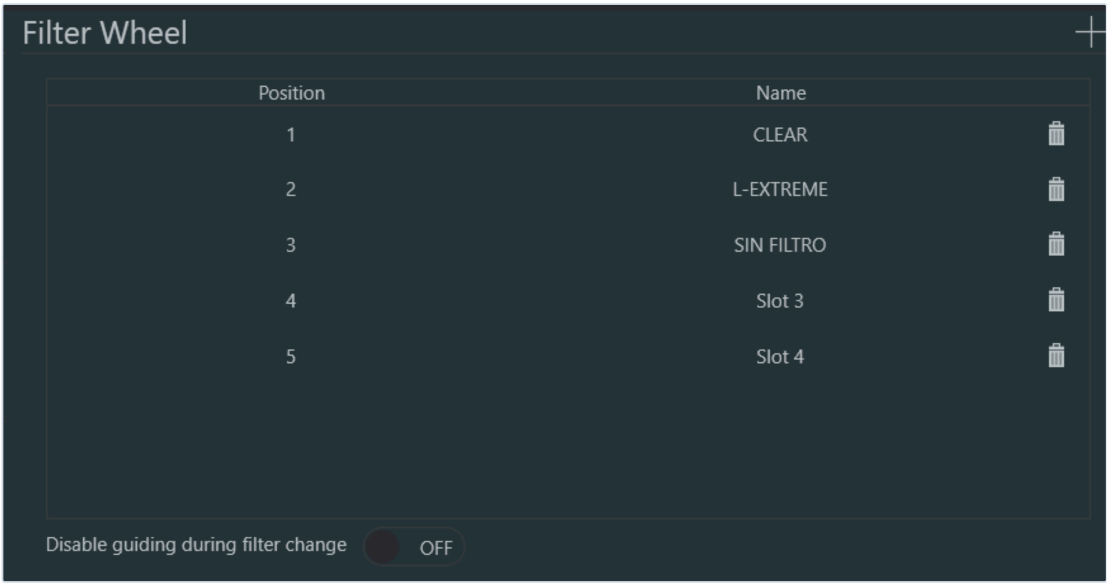
Item	Detail
Telescope	William Optics RedCat 91, 448 mm focal length, f/4.9
Mount	ZWO AM5N strain wave, no counterweight
Main camera	ZWO ASI2600MC Pro, cooled colour, 6248 x 4176 px, 3.76 um
Filter wheel	ZWO EFW 5 x 2 inch
Focuser	ZWO EAF, automatic
Guiding	ZWO ASI220 Mini on a guide scope
Flat panel	Deep Sky Dad FLAP150, motorised cover
Power	Pegasus Powerbox Advance G2
Site	Lat 31.5469 N, Lon 99.3822 W, elevation 550 m

Field of view: 3.00 by 2.01 degrees. Image scale: 1.73 arcsec per pixel.

That is a wide field. Big targets such as the North America Nebula, the Veil, the Heart and Soul, M31, the Rosette and the Pleiades all fit beautifully. Small planetary nebulae and compact galaxies will be tiny. Choose accordingly.

Filter wheel positions

Position	Label in N.I.N.A.	Use it for
1	CLEAR	Clear glass, keeps focus matched to the other slots
2	L-EXTREME	Emission nebulae, Ha and OIII dual narrowband, 7 nm
3	NO FILTER	Galaxies, star clusters, reflection nebulae, broadband



The filter wheel as configured in Options, Equipment. Position 3 is the empty slot, still labelled in Spanish.

Section 7 explains when to pick which.

2. House rules

These exist because the equipment is 8,000 km from the owner and there is nobody on site at night.

Do not touch:

• **Meridian flip settings.** Not in Options, not in the sequence. They are tuned for this mount and pier. Section 10 explains what they do and why a wrong value can put the telescope into the tripod.

• **Options, in general.** Profile, filter names, focal length, file paths, plate solving. Everything there is already correct.

• **Equipment connections.** The sequence connects and disconnects everything at the right moment. Do not press Disconnect during the night.

• **The guiding calibration.** It is done automatically.

Please do:

- Run **one target per night** on your first few sessions. Two targets means two framings, two focus runs, and twice as many ways for the night to go sideways while you sleep. One good target with 6 to 8 hours of data beats two mediocre ones.
- Leave the sequence running to the end. It closes the flat panel cover, parks the mount, warms the camera and disconnects on its own.
- Ask before improvising. A message beats a broken mount.

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3. Connecting to the observatory PC

Remote access is through **RustDesk**. You will receive the ID and the password separately, never in this document.

1. Install RustDesk on your own machine from rustdesk.com.
2. Open it, type the ID you were given into the box on the right, press Connect.
3. Enter the password when prompted.

You are now looking at the real observatory PC in Texas. N.I.N.A. is already open and, if it is not, its icon is on the desktop.

A few things about the remote desktop:

- **Never sign out, restart or shut down** the PC. If you think it needs a restart, message us instead.
- If the screen is black, move the mouse.
- A laggy connection is normal when the site link is busy. Click once and wait. Double clicking through lag is how settings get changed by accident.
- Closing the RustDesk window does not stop the sequence. The night keeps running without you.

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4. Step 1: load the sequence template

Do this before anything else. Everything that follows writes into this sequence, so it has to be open first. If you pick a target before loading the template, there is nothing for the framing step to write into.

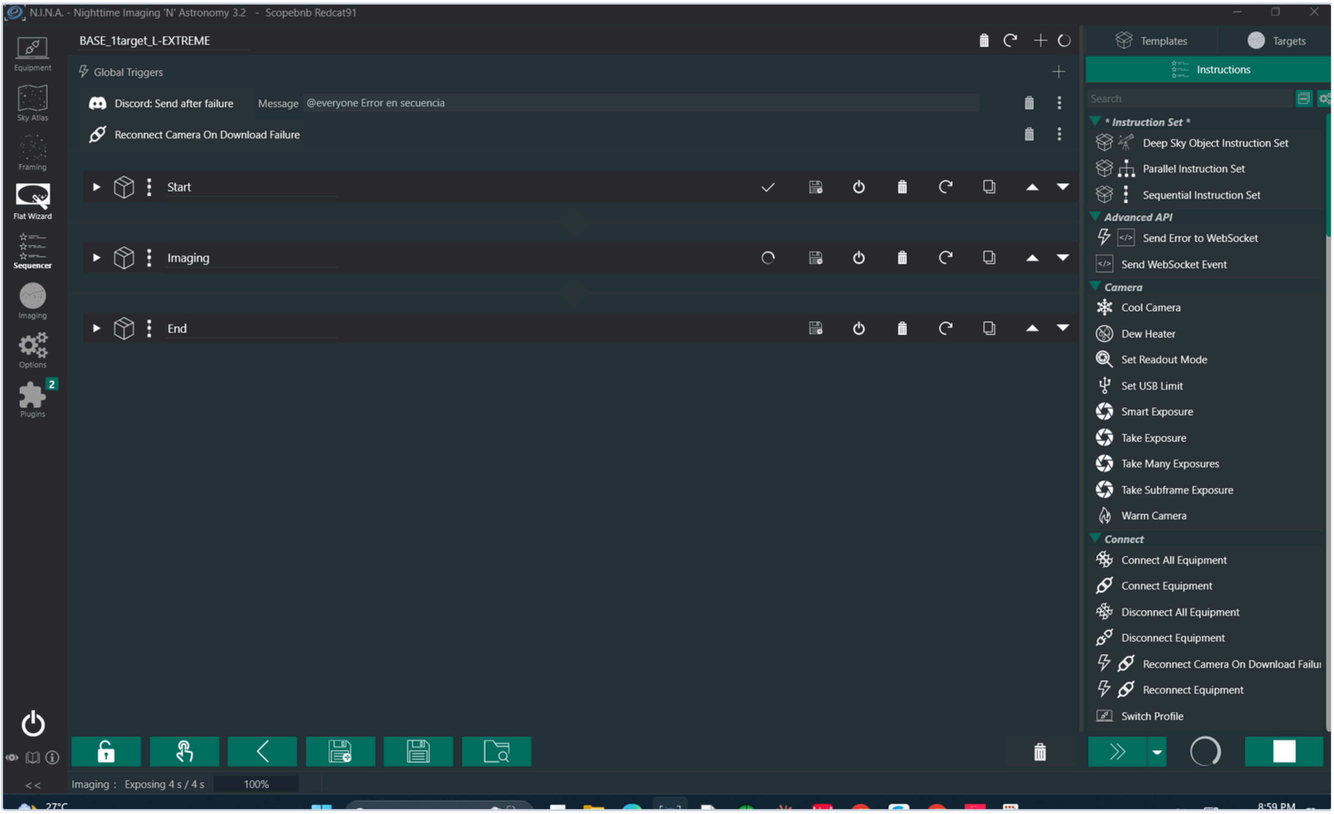
Open the **Sequencer** tab, then the **Advanced Sequencer**.

At the bottom of the sequencer there is a row of buttons. The **open folder** icon on the right loads a saved sequence. Choose:

BASE_1target_L-EXTREME or **BASE_1target_NO-FILTER**

Choose **BASE_1target_L-EXTREME** for emission nebulae, or **BASE_1target_NO-FILTER** for broadband targets such as galaxies, star clusters and reflection nebulae (see Section 7).

Both are tested templates. Each is built in three blocks:

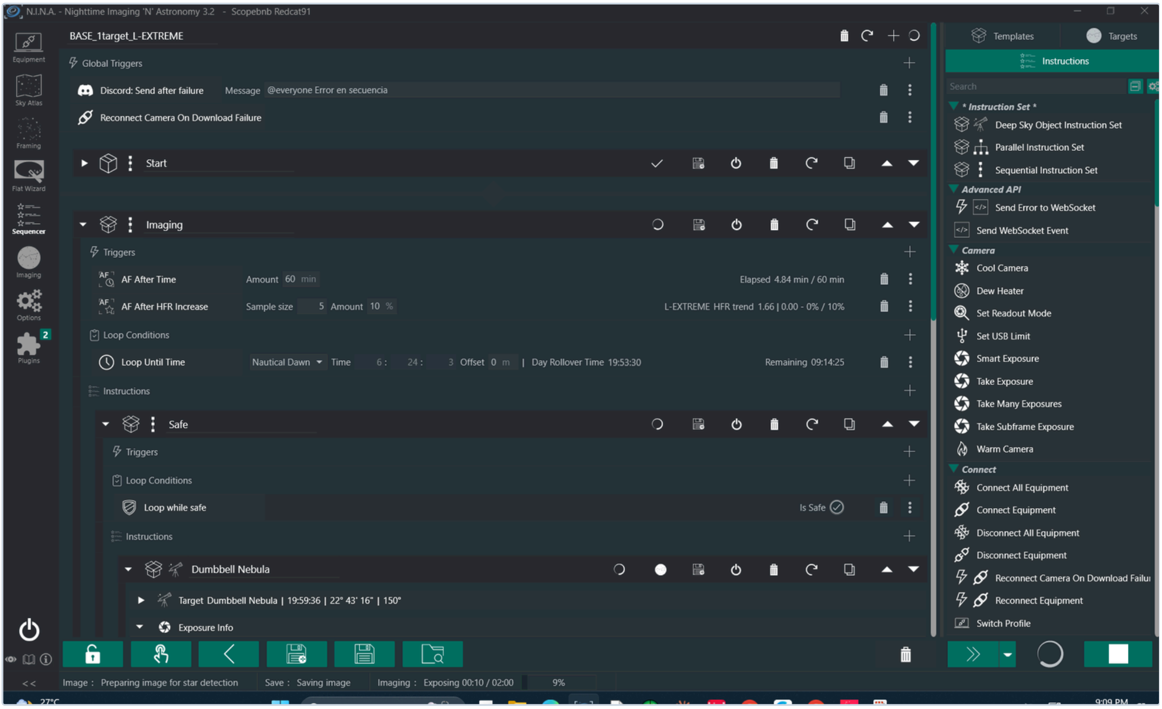


The sequence loaded and running: three blocks, Start, Imaging and End.

Start Waits until 15 minutes before nautical dusk, starts the night log, connects the safety monitor, waits until conditions are safe, connects all equipment, and starts sidereal tracking at nautical dusk.

Imaging Loops until nautical dawn, as long as conditions stay safe. Inside it sits the target: coordinates, rotation, and the exposure plan. Autofocus reruns every 60 minutes and whenever star size (HFR) grows by more than 10 percent. The target block also carries its own Meridian Flip and Center After Drift triggers.

End Stops guiding, parks the mount, closes the flat panel cover, then shoots flats and dark flats, warms the camera slowly, disconnects everything and writes the night summary.



The Imaging block open during a live session. The autofocus triggers, the loop until nautical dawn and the target itself are all visible.

The template opens with the previous night's target still in it. Replacing that target is the only edit you will make. Steps 2 and 3 do it for you.

5. Step 2: pick tonight's target

Open the **Sky Atlas** tab in the left sidebar.

You can search by name in the box at the top, but the better way to plan a night is to let N.I.N.A. tell you what is actually well placed:

1. **Date** is tonight, already filled in.

2. Set **Altitude** to **30 degrees.** Three extra fields appear.

3. **from time:** Astronomical Dusk. **through time:** Astronomical Dawn. These are already the defaults.

4. Set **visible for** to **4h.**

5. Optionally open **Object type**, **Apparent size** or **Apparent magnitude** to narrow things down. For this telescope, apparent size between 30 and 120 arcmin is the sweet spot.

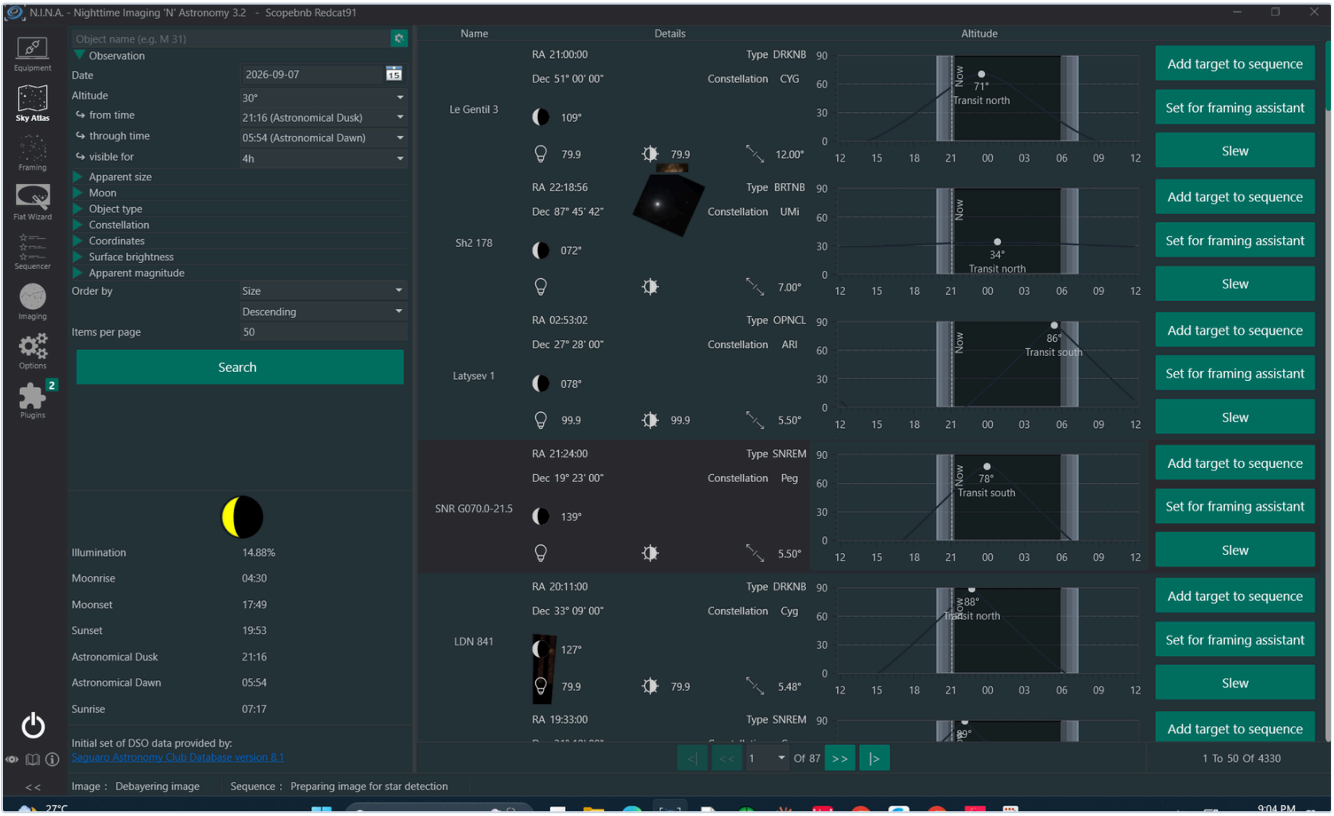
6. Press **Search.**

Each result shows the object, its coordinates, and on the right an **altitude curve** for the whole night, with a vertical "Now" band and the transit altitude marked.

The 30 degree rule

Below about 30 degrees you are shooting through two or more air masses. Stars bloat, colour smears, and gradients from the horizon creep in. So:

- The target must stay **above 30 degrees for at least 4 hours** inside the dark window.
- Prefer targets that **transit near the middle of the night**. A target that transits at 20:30 is already falling when you start.
- If the curve shows a narrow hump only just clearing 30 degrees, **pick another target**. There is always something better placed. That is exactly what the search filter you just set is for.



Sky Atlas with the altitude filter set to 30 degrees for 4 hours. Each row shows the object, its coordinates and its altitude curve for the night, with the current time marked.

Once you have chosen, press **Set for framing assistant** on that row. This carries the coordinates into the next step.

> There is also a **Slew** button on each row. Do not use it. The sequence handles all slewing.

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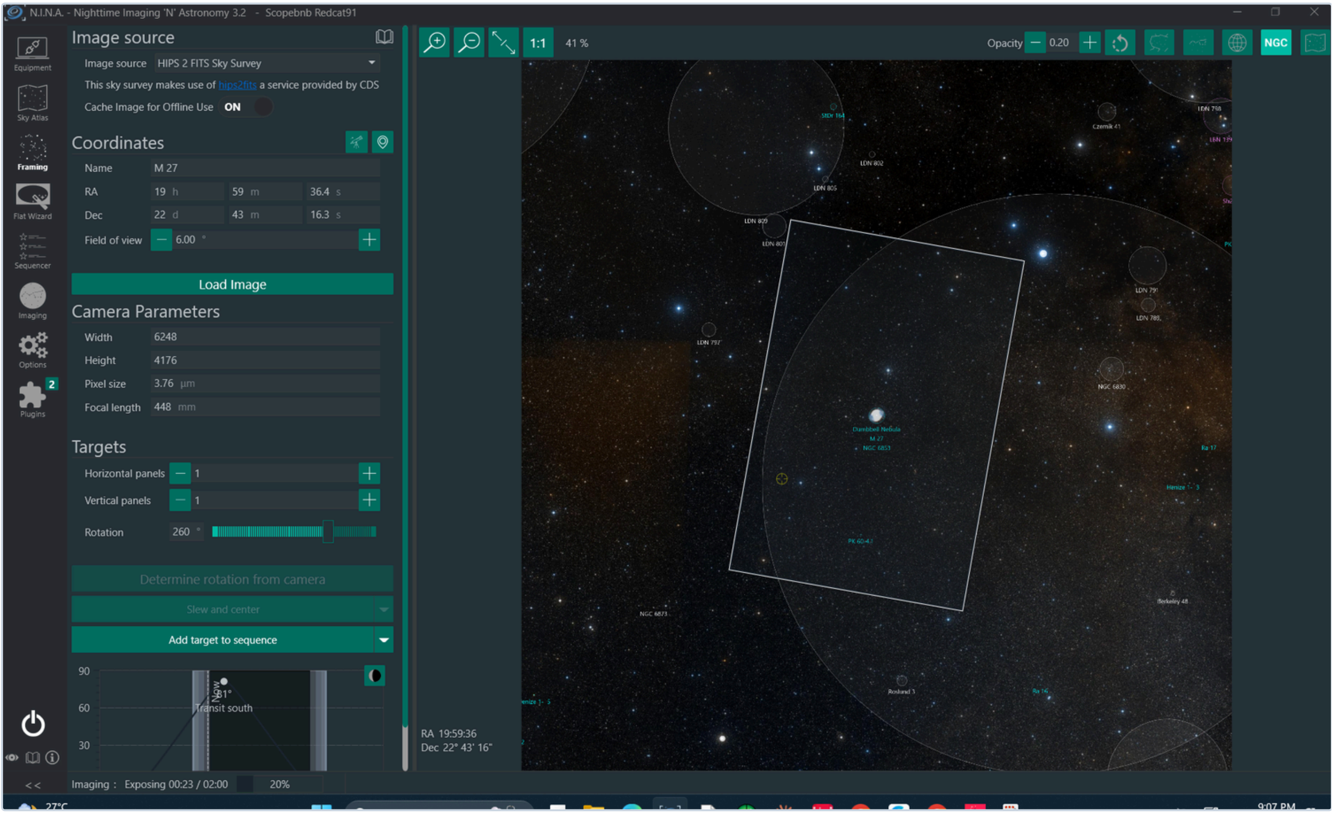
6. Step 3: frame it and write it into the sequence

You land in the **Framing** tab with the object already loaded.

1. Press **Load Image**. N.I.N.A. downloads a sky survey image of the region from the HIPS 2 FITS service. This takes 10 to 30 seconds.

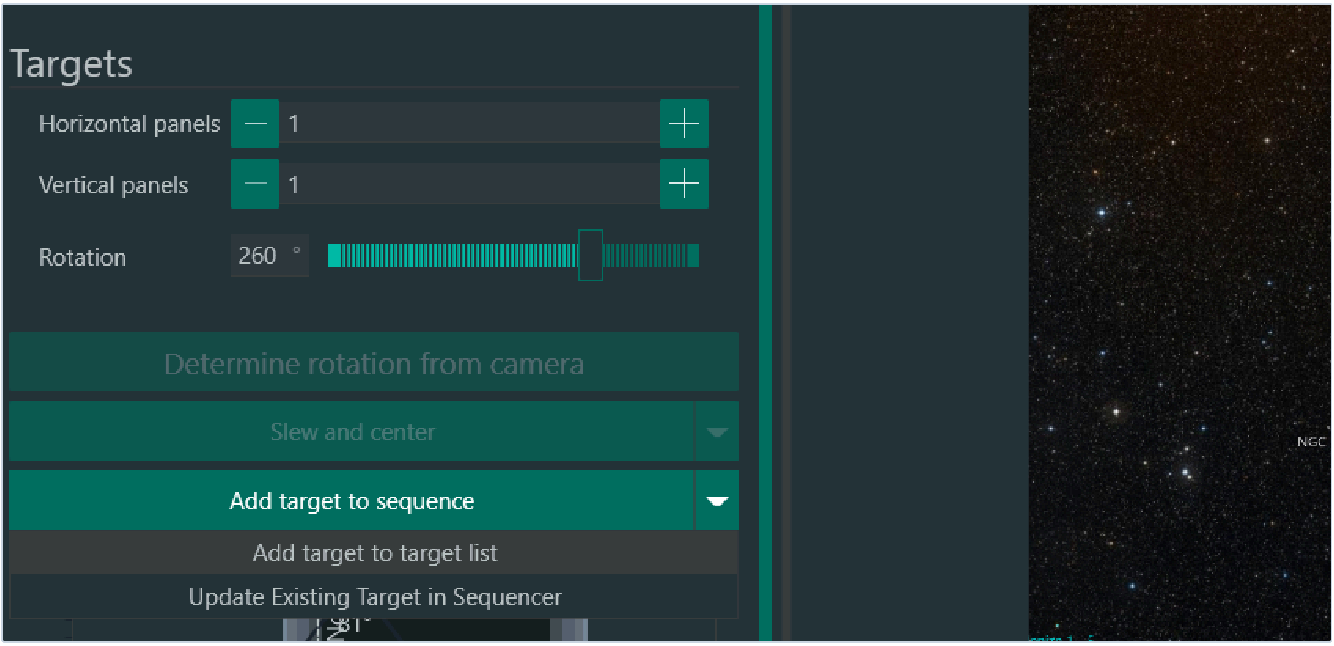
2. A white rectangle appears over the image. That rectangle is exactly what the camera will see.
3. **Drag the rectangle** to place your object where you want it. Off centre is often better than dead centre, especially for a target with a companion or a nice star field on one side.
4. Adjust **Rotation** with the slider or by typing degrees. Rotation is what turns an awkward crop into a good composition. The rig has no field rotator, so this angle is achieved by the framing itself, and the sequence plate solves to reach it.
5. Leave **Horizontal panels** and **Vertical panels** at **1**. Mosaics are not for a first night.



The Framing tab with the survey image loaded. The white rectangle is the exact field the camera will record.

When you are happy, use the small **arrow** on the right of the **Add target to sequence** button and choose:

Update Existing Target in Sequencer



The dropdown next to Add target to sequence. Update Existing Target in Sequencer is the one you want.

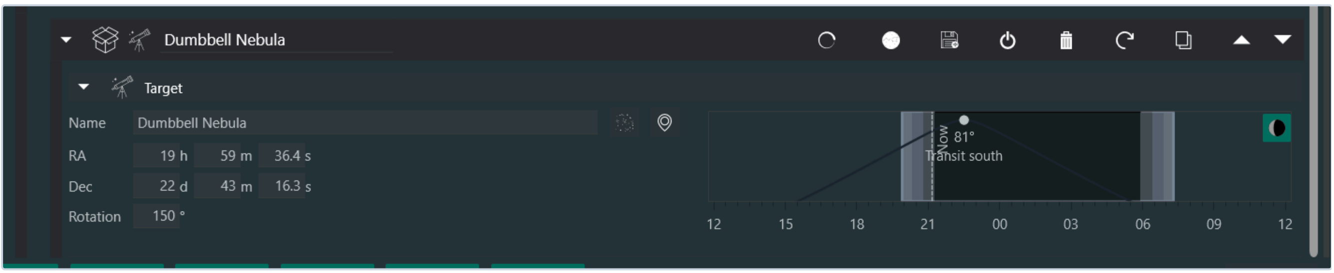
That writes the name, RA, Dec and rotation into the sequence you loaded in step 1, replacing the old target and leaving everything else in the sequence untouched. This is why the sequence had to be loaded first: without it there is no existing target to update.

> If you press the main **Add target to sequence** button instead, you get a second target added and the sequence will try to shoot both. Use the dropdown.

Check the result before you start

Go back to the **Sequencer** tab and open the target block inside **Imaging**. It should now read:

- **Name**: your object
- **RA**, **Dec**: hours, minutes, seconds and degrees, minutes, seconds
- **Rotation**: the angle you set in Framing

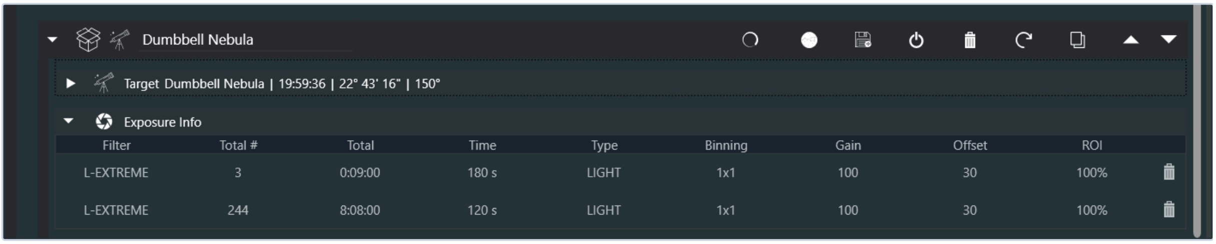


The target block. These four fields are the only thing that changes from night to night.

You can also type these four fields by hand right here. That is the fallback if the framing step did not take.

Then open **Exposure Info**. The template ships with:

Filter	Frames	Exposure	Type	Binning	Gain	Offset
L-EXTREME	3	180 s	LIGHT	1x1	100	30
L-EXTREME	242	120 s	LIGHT	1x1	100	30



Exposure Info in the loaded template.

The large frame count is deliberate. The sequence stops at nautical dawn, so it simply takes as many frames as the night allows. Leave the count alone. What you may change is the **filter** and, if you have a reason, the **exposure time**.

Everything else in the sequence stays exactly as it is.

7. Step 4: L-EXTREME or NO FILTER

This is the single most important creative decision of the night, and the most common beginner mistake is using the narrowband filter on a target that has no emission lines.

Use L-EXTREME (position 2) for

Emission nebulae. Anything that glows in hydrogen alpha or doubly ionised oxygen:

• North America and Pelican, Veil, Heart and Soul, Rosette, Elephant’s Trunk, Crescent, California

• The Lagoon, Eagle, Trifid and Omega in summer

• Most objects with “nebula” in the name that are not reflection nebulae

The filter passes two narrow 7 nm windows and blocks everything else, so it kills light pollution and, importantly, **works under a bright Moon.** If the Moon is up and more than half illuminated, this is your filter.

Typical exposure: **120 to 300 seconds**, gain 100.

Use NO FILTER (position 3) for

Anything whose light is broadband, meaning a continuous spectrum rather than emission lines:

• Galaxies: M31, M33, M81 and M82, the Leo Triplet

• Star clusters: the Pleiades, the Double Cluster, M11

• Reflection nebulae: the blue dust around the Pleiades, the Iris Nebula

• Comets

- Broad dusty fields, integrated flux nebulae, dark nebulae

Putting the L-eXtreme on a galaxy throws away roughly 95 percent of its light and returns a red, lifeless image. Do not do it.

Typical exposure: **60 to 120 seconds**, gain 100. Broadband saturates faster, and the sky background builds up quickly, so shorter subs and more of them.

The Moon test

Check the Moon panel at the bottom left of Sky Atlas before you commit. It shows illumination, moonrise and moonset.

- Moon below the horizon all night: either filter works. Broadband targets are at their best.
- Moon up and less than 40 percent: broadband is still usable if the target is far from the Moon.
- Moon up and more than 40 percent: **use L-EXTREME and pick an emission nebula.** A broadband night under a gibbous Moon is a wasted night.

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8. Step 5: start the sequence

Press the **start** button at the top of the sequence.

Then leave it alone. From here everything is automatic:

- It waits for dusk. Nothing appears to happen for a while. That is correct.
- It connects the equipment, waits for the safety monitor to report safe, starts tracking.
- It slews, plate solves, rotates the framing to match, and centres.
- It runs an autofocus, starts guiding, and begins exposing.
- It refocuses every hour and whenever the stars grow.
- It flips at the meridian, recentres, refocuses, resumes.
- At nautical dawn it shoots calibration frames, parks, closes the cover, warms the camera and shuts down.

Progress messages are also sent to the SCOPEbnb Discord, including a message if something fails.

If clouds arrive, the safety monitor pauses the sequence. When it clears, the sequence resumes on its own. This is normal and you do not need to intervene.

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9. Step 6: watching the night

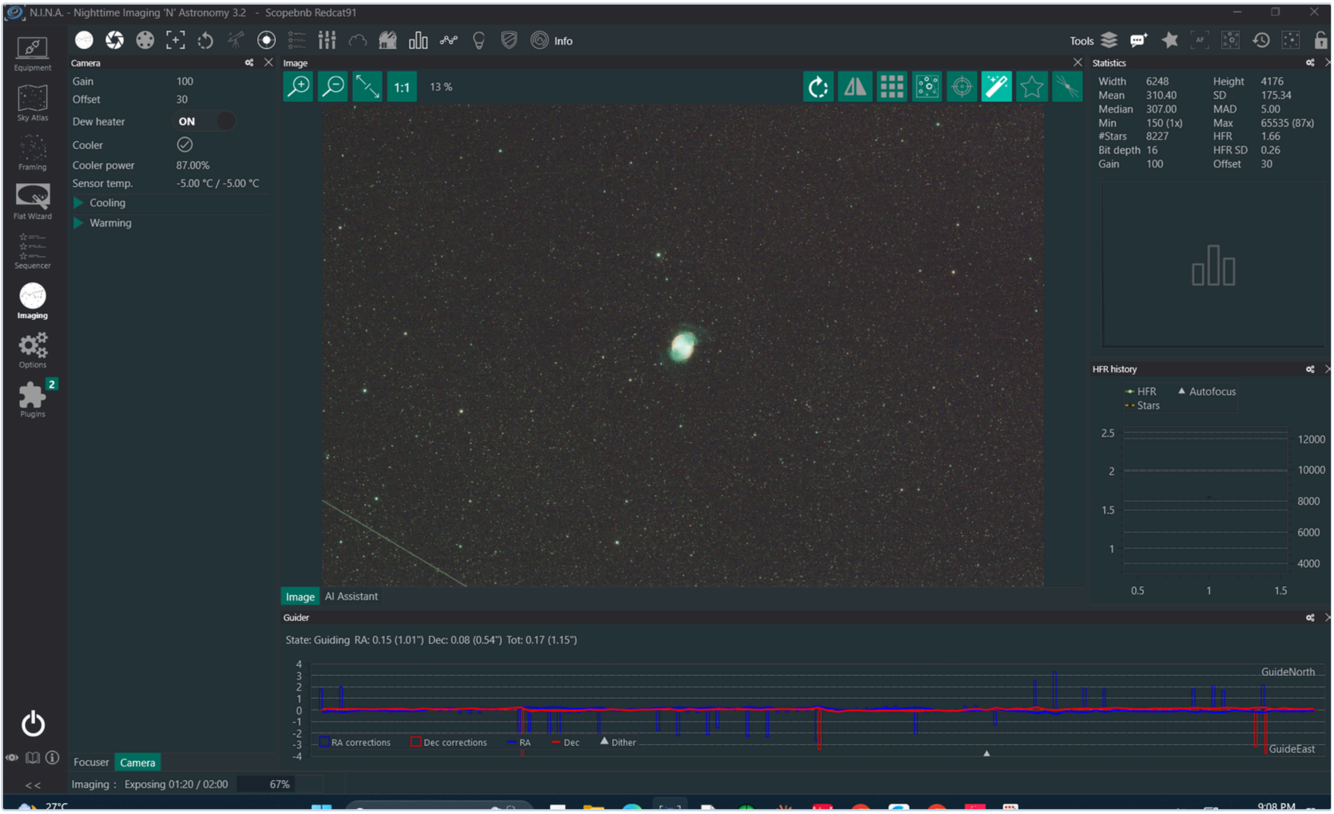
Switch to the **Imaging** tab. Four things are worth looking at.

The image panel. Each frame appears here as it is downloaded. Do not judge colour or brightness on a single 120 second sub, it is meant to look flat and dull. Judge the stars.

HFR history, top right. HFR is half flux radius, essentially star size in pixels. A flat line means focus and seeing are stable. A slow upward drift means the tube is cooling and autofocus will correct it at the next trigger. A sudden spike usually means cloud.

Statistics, above it. #Stars and HFR per frame. A large drop in star count means cloud passing.

Guider graph, bottom. RA and Dec corrections in arcseconds. On a good night this rides between plus and minus 1 arcsecond. Occasional spikes are wind or a strain wave periodic error and are normal. A sustained runaway in one axis is worth reporting.



A real session on this rig: the Dumbbell Nebula on a single 120 second sub through the L-eXtreme, 8227 stars detected, HFR 1.66, and guiding at 1.16 arcsec total.

You can close the remote desktop at any point. The night continues without you.

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10. Meridian flip: what it is and why you must not touch it

The mount tracks your target from east to west. When the target crosses the meridian, the imaginary line running north to south overhead, the telescope would eventually collide with the mount or the pier if it kept going. So the mount rotates 180 degrees in both axes and picks the target up from the other side. That is a meridian flip.

N.I.N.A. handles it: it pauses, flips, plate solves, recentres to your exact framing, refocuses, restarts guiding and carries on. You lose one or two frames and nothing else.

The settings that govern it are minutes after meridian, maximum minutes after meridian, use telescope side of pier, recenter after flip, scope settle time and pause before meridian. **Do not change any of them, in Options or in the sequence.** They are matched to this specific pier height, this tube length and this mount's limits. A value that is too large lets the tube swing into the pier in the dark, with nobody there to stop it. A value that is too small makes the mount flip and unflip repeatedly and ruins the night.

If a flip fails, the sequence stops safely and the Discord alert fires. Leave it and tell us.

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11. Pier camera and time lapse

There is a second, small camera watching the pier. It is useful for seeing whether the sky is actually clear, whether the cover really opened, and for making a nice time lapse of your night.

1. On the observatory desktop, open **ASIStudio** and choose **ASICap**. There is also a direct ASICap shortcut.
2. Connect the pier camera from the camera dropdown at the top.
3. You now have a live view. Exposure and gain controls are on the left panel. At night, raise the exposure until you can see stars and the silhouette of the telescope.

To record a time lapse:

1. In the record settings, choose an **interval** between frames. 10 to 20 seconds gives a pleasant result over a full night.
2. Set the output format and the destination folder.
3. Press **record** to start, and **stop** when you are done.

Please stop the recording before you leave for the night if you set a long one, and tell us if you leave a recording running, so that the disk does not fill up.

The pier camera is independent of the imaging sequence. Using it does not disturb anything.

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12. Where your files go

Light frames are saved as **FITS** to the observatory's Synology drive, sorted automatically:

```
<date>\LIGHT\<datetime>_<target>_<filter>_<sensor
temp>_<exposure>s_rot<angle>_foc<position>_<frame number>.fits
```

Calibration frames from the end of the sequence land in the same tree. A **night report** is written for every session, listing frames taken, HFR, guiding statistics and any errors.

You will be told separately how your night's folder is shared with you.

A note on colour: the ASI2600MC Pro is a one shot colour camera with an RGGB Bayer matrix. Your FITS files are undebayered. Your processing software needs to debayer them as RGGB. In Siril, PixInsight or APP this is a single setting.

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13. Quick reference

I want to	Where
Load the night's sequence (do this first)	Sequencer, Advanced Sequencer, open folder: **BASE_1target_L-EXTREME** or **BASE_1target_NO-FILTER**
Find a target that is well placed	Sky Atlas, Altitude 30 degrees, visible for 4h
Compose the shot	Framing, Load Image, drag and rotate
Write it into the sequence	Framing, dropdown, Update Existing Target in Sequencer
Change the filter	Sequence, target block, Exposure Info
Watch the night	Imaging tab
See the sky and the rig	ASICap

If something looks wrong

Symptom	What it usually is	What to do
Nothing happens after start	Waiting for dusk or for the safety monitor	Nothing. Check the time in the Start block
Sequence paused	Clouds, safety monitor unsafe	Nothing. It resumes on its own
Stars growing, HFR climbing	Tube cooling, or cloud	Nothing. Autofocus will fire
Guiding graph spiking constantly	Wind, or a passing cloud	Nothing for a few minutes. Report if it persists
Sequence stopped with an error	Various	Do not restart anything. Contact us
Frames look red and empty	L-eXtreme on a broadband target	Stop, change the filter to NO FILTER, restart

Contact

Message us on Discord for anything at all. There is no such thing as a question that is too basic on a first night, and we would far rather answer one than have you guess.

Clear skies.

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